Project 1: HR data cleaning and analysis

Final Project

January 2022 – January 2023 | 8,049 raw records → 7,787 analysis-ready records

1. Executive Summary

This project applies a complete SQL-based data wrangling pipeline to a real-world construction company payroll dataset, demonstrating skills in data integration, cleaning, standardization, deduplication, and descriptive analysis. The source data was distributed across four relational tables — salaries, companies, employees, and functions and required resolution of twelve distinct data quality issues before any analysis could be performed.

The raw dataset contained 8,049 payroll records spanning 13 months and 24 companies. After systematic cleaning, 7,787 valid records remained, representing 642 active employees across three Brazilian cities. All transformations were performed on a working copy of the data, leaving original source tables untouched. The final analysis reveals a growing workforce, a clear salary hierarchy tied to job function, and a geographically concentrated employee base centered in Goiania.

2. Dataset Overview

The dataset originates from DM Company, a Brazilian construction holding firm operating residential, commercial, and hospital construction sites across the states of Goiás, Distrito Federal, and Tocantins. The four source tables and their roles in the data model are described below:

Table	Rows	Key Column	Description
salaries	8,049	employee_id	Monthly payroll records — the primary fact table containing salary, date, company, and function per employee per month
companies	25	company_name	Company metadata including city, state, company type, and construction site category
functions	108	function_code	Job function codes mapped to human-readable titles and broader function groups
employees	1,156	employee_code_emp	Employee demographics: name, gender, age, and company code

A LEFT JOIN across all four tables on their respective key columns produced the emp_dataset staging table, from which a refined df_employee working table was built containing only the 13 columns relevant to analysis.

3. Data Quality Issues Identified

A systematic audit was performed across all columns in all four tables before any transformations were applied. Twelve distinct issues were identified, spanning missing values, duplicate records, invalid test data, typographical errors, casing inconsistencies, encoding problems, and structural/type mismatches:

Category	Issue	Detail	Resolution
Missing Values	Null salary	70 rows with no salary value	Deleted — cannot be imputed
Missing Values	Null site category	4 admin office rows with no category	Deleted — not valid site records
Duplicates	Repeated payroll rows	186 rows sharing same employee + month	Removed via CTE + ROW_NUMBER()
Invalid Data	HR test entries	8 rows with salary = R$1,000,000	Deleted — confirmed system tests
Typos	City misspelling	'Goianiaa' (extra 'a')	Corrected to 'Goiania'
Typos	Category misspelling	'Commerciall' (extra 'l')	Corrected to 'Commercial'
Typos	Company type variants	'Construction Sites', 'Construction Site s'	Normalized to 'Construction Site'
Casing	State all-caps	'GOIAS'	Standardized to 'Goias'
Encoding	Invalid column name	GEN(M_F) parsed as a formula	Renamed to 'gender' via sp_rename
Standardization	Gender abbreviations	'M' / 'F' single characters	Expanded to 'Male' / 'Female'
Data Types	Timestamp date column	Date stored with irrelevant time component	Cast to DATE; truncated to YYYY-MM
Whitespace	Padding in text fields	Leading/trailing spaces across all columns	TRIM() applied in single UPDATE

4. SQL Techniques & Skills Demonstrated

The following SQL features were used across the data cleaning and analysis pipeline. Each technique served a specific purpose in transforming the raw source data into the final analysis-ready table:

- **Joins:** Four-table LEFT JOIN on matching key columns merged all source tables into a single staging dataset without dropping records that lacked a join match, which helped surface referential integrity gaps early.
- **Temporary Tables:** Two intermediate tables were created using SELECT INTO emp_dataset for the raw join result and df_employee for the refined working table. This non-destructive approach preserved all original source data throughout the project.
- **CTEs & Window Functions:** A CTE (rncte) assigned ROW_NUMBER() partitioned by employee_id and pay_month to flag duplicate rows. A DELETE WHERE row_num > 1 then removed all but the first occurrence of each duplicate group. SUM() OVER () calculated city percentage shares without a self-join.

- **Aggregate Functions:** COUNT(DISTINCT employee_id), AVG(salary), ROUND(), TOP(1) with ORDER BY, and GROUP BY combinations were used throughout the analysis section to compute all descriptive statistics.
- **String Functions:** TRIM() removed whitespace from all text columns in one UPDATE statement. LEFT() truncated dates to YYYY-MM. CONCAT() built the composite unique identifier. CASE WHEN standardized gender values.
- **Data Type Conversion:** CAST(date AS DATE) converted timestamp strings to proper date types. ALTER TABLE added a computed pay_month column derived via LEFT(CAST(month_year AS VARCHAR), 7) for consistent monthly grouping.
- **Dynamic Subqueries:** A correlated subquery SELECT MAX(pay_month) FROM df_employee was reused across all analysis queries to dynamically target the most recent pay period, making queries resilient to future data refreshes without hardcoded dates.

5. Analysis Results & Descriptive Statistics

5.1 Has Average Salary Increased or Decreased Since January 2022?

The dataset covers January 2022 through January 2023, so a comparison beyond that point is not possible with available data. Within the dataset, average salary declined steadily from R$1,666.12 in January 2022 down to a low of R$1,573.61 in September 2022, before partially recovering to R$1,581.28 by January 2023.

Month	Avg Salary	Month	Avg Salary	Month	Avg Salary
Jan 2022	R$1,666.12	May 2022	R$1,604.33	Sep 2022	R$1,573.61
Feb 2022	R$1,652.58	Jun 2022	R$1,592.89	Oct 2022	R$1,599.49
Mar 2022	R$1,638.52	Jul 2022	R$1,581.96	Nov 2022	R$1,590.42
Apr 2022	R$1,616.03	Aug 2022	R$1,581.92	Dec 2022 / Jan 2023	R$1,574.79 / R$1,581.28

5.2 Highest & Lowest Average Salary by Function Group

Engineering is the highest-paid function group at R$3,208.82, followed by Production Supervisors at R$2,832.84. Trainees are the lowest paid at a flat R$950.00 less than a third of the Engineering average. Professionals and Assistants make up 78% of total headcount but both earn below the company-wide average of R$1,638.

Function Group	Average Salary	vs. Overall Avg
Engineering ← Highest	R$3,208.82	+96%
Production Supervisors	R$2,832.84	+73%
Administration	R$1,974.37	+21%
Machine Operators	R$1,753.19	+7%
Professionals	R$1,699.61	+4%

Function Group	Average Salary	vs. Overall Avg
Assistants	R$1,217.58	-26%
Trainees ← Lowest	R$950.00	-42%

5.3 Gender Pay Gap by Company & Function Group

The Villas at Hunter's Glen has the largest company-level gender pay gap: male employees earn R$2,103.00 on average versus R$950.00 for female employees a 121% gap. This is driven by role composition (men in Engineering, women in Trainee roles) rather than direct pay inequality. Notably, Regional Hospital is an outlier where women out-earn men by R$121.97 (8.1%).

Company	Female Avg	Male Avg	Gap %
The Villas at Hunter's Glen ← Largest	R$950.00	R$2,103.00	121.4%
The Meadows at Sunset Ridge	R$1,339.45	R$1,737.69	29.7%
The Crossings at Falcon Point	R$1,370.00	R$1,647.43	20.3%
The Glades at Maplewood	R$1,319.86	R$1,582.82	19.9%
Regional Hospital (F > M)	R$1,620.23	R$1,498.26	8.1%

By function group, gaps are small across the board. Production Supervisors show the largest gap at 6.0% in favor of women. Engineering has no female representation in the dataset. All other groups are within 6%, with Assistants and Trainees essentially equal.

Function Group	Female Avg	Male Avg	Gap %	Higher Earner
Production Supervisors	R$2,990.33	R$2,819.98	6.0%	Female
Administration	R$1,885.47	R$1,989.19	5.5%	Male
Machine Operators	R$1,679.03	R$1,758.16	4.7%	Male
Professionals	R$1,741.61	R$1,692.83	2.9%	Female
Assistants	R$1,218.47	R$1,217.40	0.1%	Female
Trainees	R$950.00	R$950.00	0.0%	Equal
Engineering	No data	R$3,208.82	—	Male only

5.4 Salary Distribution Across States

Distrito Federal (Brasilia) has the highest average salary at R$1,649.26, followed by Tocantins (Palmas) at R$1,626.54. Goiás (Goiania) has the lowest average at R$1,567.67 despite hosting 63.9% of all employees. The median is a sharper indicator: Brasilia's median salary of R$1,683 is 36% higher than Goiania's R$1,239, reflecting a greater concentration of mid-to-senior roles at Brasilia sites.

State	City	Records	Avg Salary	Median Salary	Min	Max
Distrito Federal	Brasilia	2,632	R$1,649.26	R$1,683.00	R$950	R$5,500
Tocantins	Palmas	290	R$1,626.54	R$1,290.45	R$950	R$3,885
Goiás	Goiania	3,885	R$1,567.67	R$1,239.33	R$950	R$5,500

6. Conclusions

This project successfully demonstrated the end-to-end SQL data wrangling lifecycles, from raw, multi-table relational data to a fully cleaned, standardized, and analysis-ready dataset. Twelve distinct data quality issues were identified through systematic auditing and resolved using targeted SQL techniques, all without modifying any original source tables.

The four key analytical questions reveal important workforce patterns: average salary declined 5.1% over the dataset period as rapid headcount growth brought in more lower-wage workers; Engineering is the highest-paid function group (R$3,208.82) and Trainees the lowest (R$950.00); The Villas at Hunter's Glen has the largest company-level gender pay gap (121%) driven by role composition; and Brasilia employees earn a meaningfully higher median salary (R$1,683) than their Goiania counterparts (R$1,239) despite Goiania hosting 64% of the workforce.

Key SQL skills applied include multi-table joins, non-destructive temp table creation, CTE-based deduplication with window functions, aggregate analysis with dynamic subqueries, and comprehensive string and type standardization. All queries are fully documented with inline comments and result annotations in the accompanying final_project.sql file.